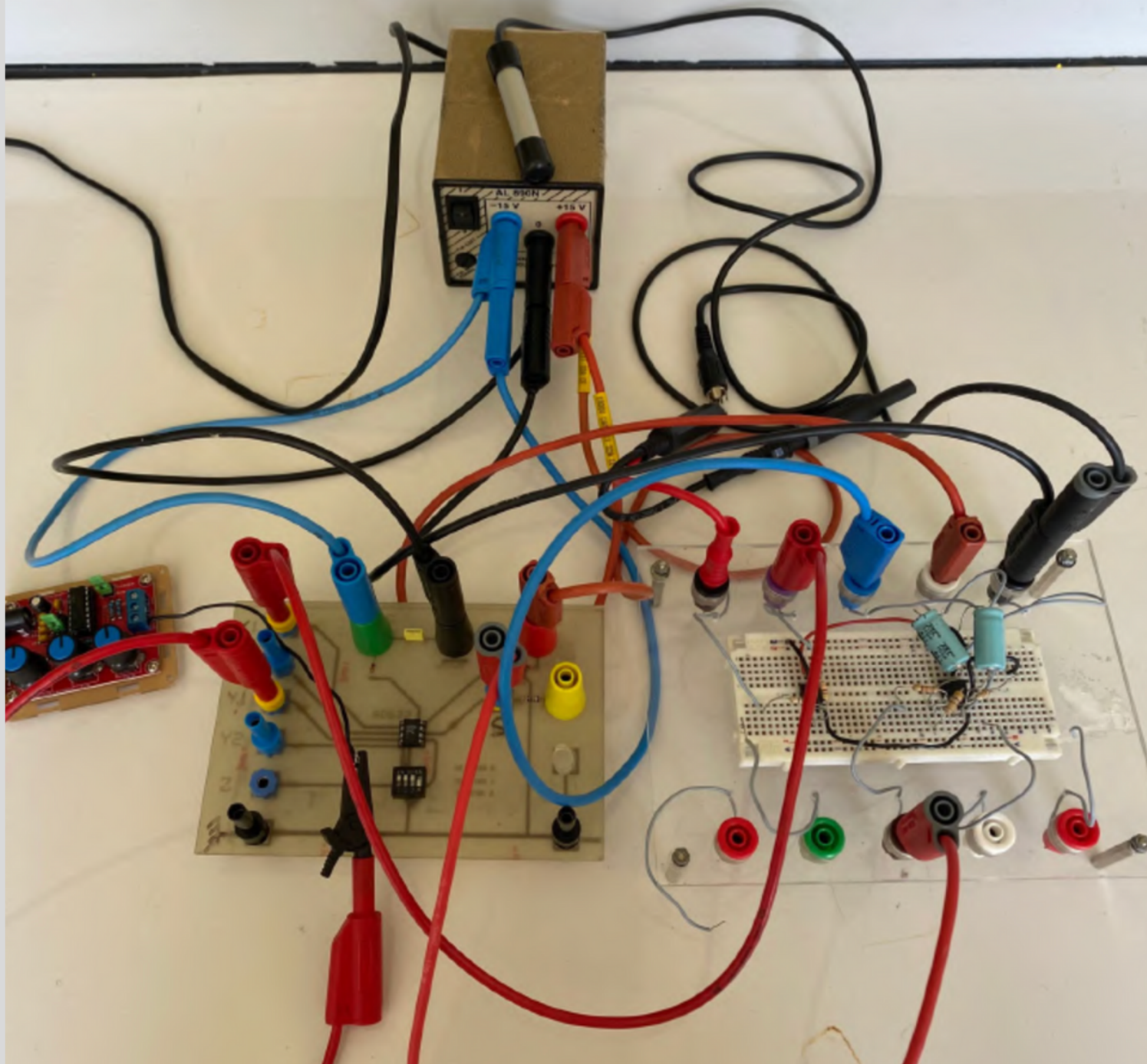


# Wind Noise Reduction in Ski Helmets using Lock-In Amplifier



## Methodology

I used analog components such as a function generator, multiplier, RC filters, and oscilloscope to construct a synchronous detection system.



The incoming signal from a microphone was multiplied by a reference sine wave (same frequency), then passed through a low-pass filter to remove noise.

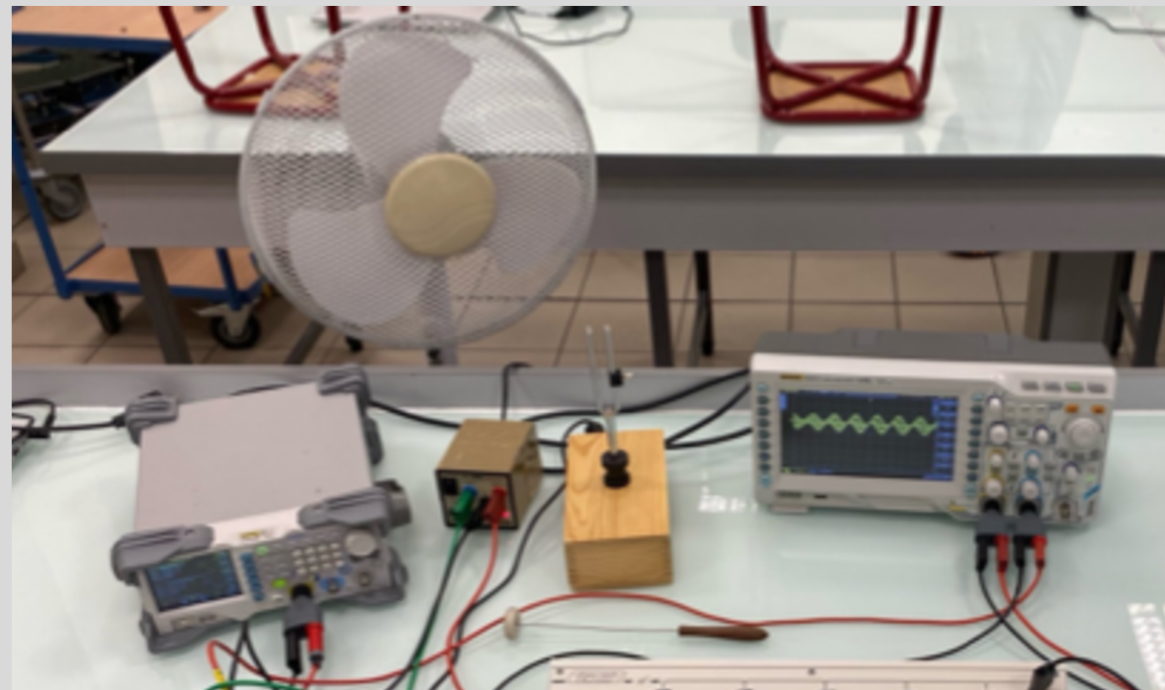


Several versions of the filter were tested (passive first-order and active second-order) to compare efficiency.



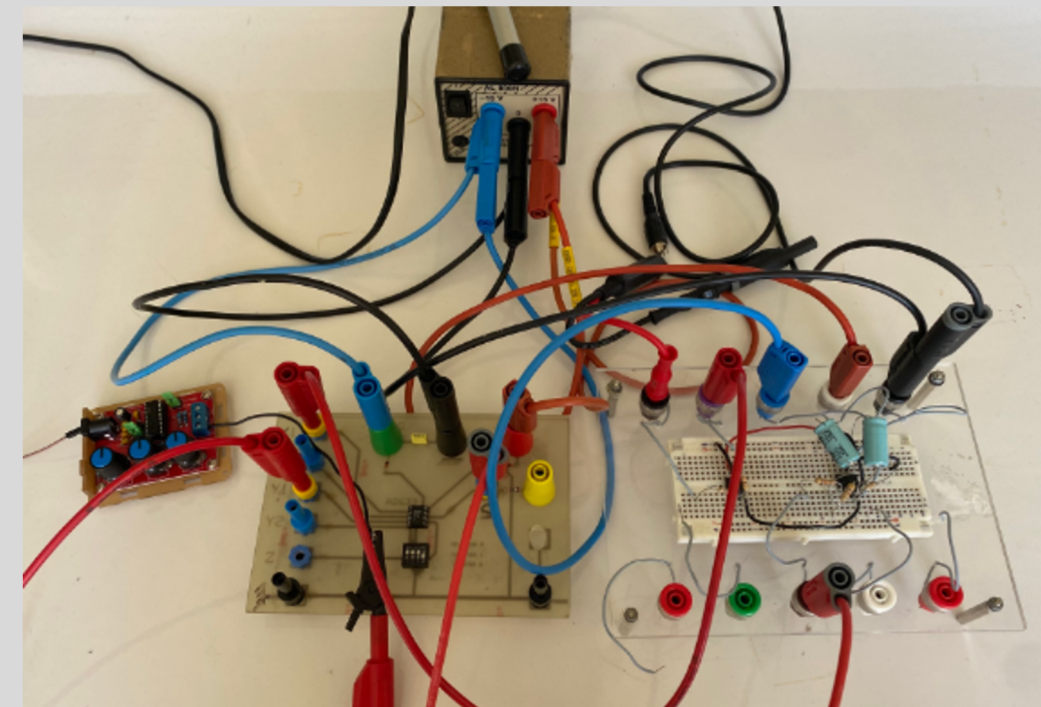
# Experiments

I first carried out tests using a tuning fork to generate stable, low-frequency signals, allowing me to validate the basic operation of the synchronous detection circuit in ideal conditions.



To simulate wind noise, I used a household fan directed at the microphone, enabling me to observe how the system responded to real, fluctuating noise.

Finally, I built a miniaturized version of the setup using compact analog components on a breadboard, to assess its practicality for integration into a ski helmet.





# Results:

The active second-order low-pass filter provided significantly better attenuation of high-frequency noise compared to the passive first-order design, making the output signal much cleaner.

| Frequency | U <sub>e</sub> (V) | U <sub>s</sub> (V) | % of the amplitude of the wave passing through the filter |
|-----------|--------------------|--------------------|-----------------------------------------------------------|
| 100Hz     | 5                  | 4,85               | 97                                                        |
| 200Hz     | 5                  | 4,5                | 90                                                        |
| 300Hz     | 5                  | 4,1                | 82                                                        |
| 400Hz     | 5                  | 3,7                | 74                                                        |
| 500Hz     | 5                  | 3,4                | 68                                                        |
| 600Hz     | 5                  | 3,2                | 64                                                        |
| 700Hz     | 5                  | 2,9                | 58                                                        |

*First order passive low – pass filter*

| Frequency | U <sub>e</sub> (V) | U <sub>s</sub> (V) | % of the amplitude of the wave passing through the filter |
|-----------|--------------------|--------------------|-----------------------------------------------------------|
| 100Hz     | 5                  | 4,1                | 82                                                        |
| 200Hz     | 5                  | 3,1                | 62                                                        |
| 300Hz     | 5                  | 2,1                | 42                                                        |
| 400Hz     | 5                  | 1,6                | 32                                                        |
| 500Hz     | 5                  | 1,3                | 26                                                        |
| 600Hz     | 5                  | 1                  | 20                                                        |
| 700Hz     | 5                  | 0,8                | 16                                                        |

*Second order active low – pass filter*

The overall system proved effective at isolating low-frequency signals even in noisy environments, though further tuning and optimization could enhance its selectivity and robustness.